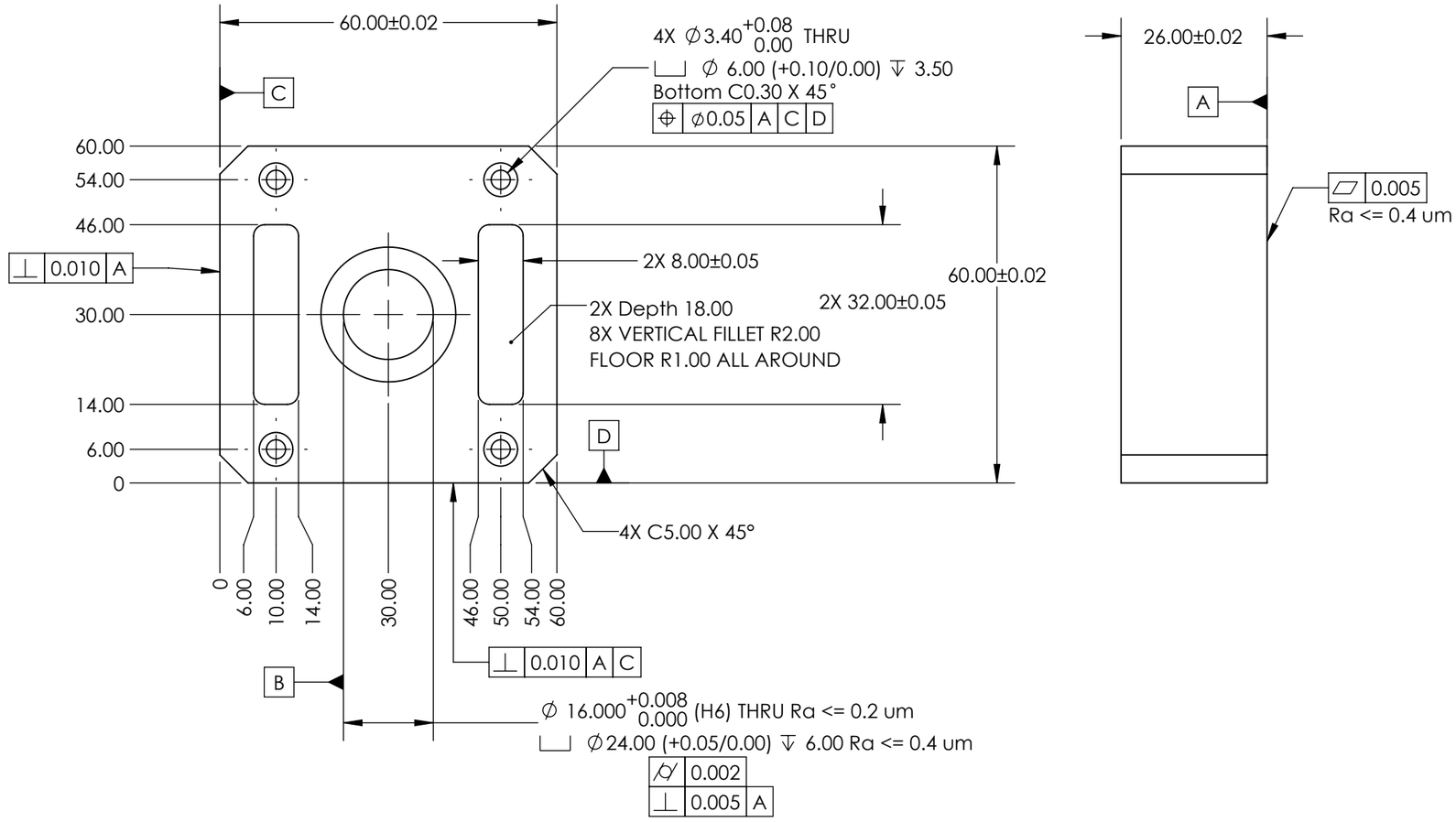
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

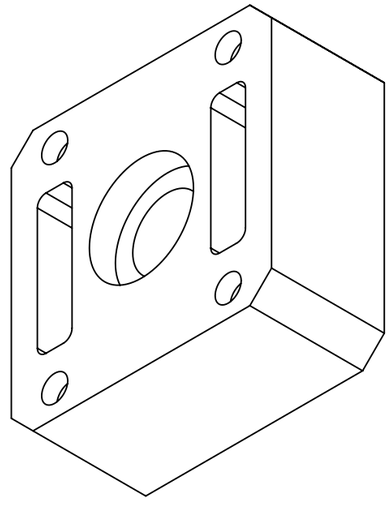
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-SEMI-FLIPCHIP-BONDHEAD-6.4.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/-0.02 mm, ANGULAR +/-0.1 DEG (ULTRA-PRECISION 5-AXIS TURN-MILLING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

2. SEMICONDUCTOR FLIP-CHIP & HIGH-ACCELERATION (20G) RELIABILITY:
2.1 MATERIAL OPTION A: VACUUM-MELTED TITANIUM ALLOY Ti-6Al-4V ELI (GRADE 23 / AMS 4930), ANNEALED & STABILIZED, HIGH SPECIFIC STIFFNESS.
2.2 MATERIAL OPTION B: FORGED AEROSPACE ALUMINUM 7075-T6 (AMS 4045).
2.3 100% SUBTRACTIVE 5-AXIS CNC MACHINING ONLY. NO ADDITIVE MANUFACTURING (SLM/SLS) OR POWDER SINTERING PERMITTED TO AVOID DIE CONTAMINATION AND MICRO-CRACKING UNDER HIGH-G IMPACT.

3. KINEMATIC RIGIDITY & ULTRA-HIGH COPLANARITY GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (VCM / LOAD-CELL MATING FACE): FLATNESS <= 0.005 mm, SURFACE FINISH Ra <= 0.4 MICRONS.
3.2 SECONDARY DATUM [B] (COLLET GUIDE BORE Ø16.000 +0.008/0.000 H6): CYLINDRICITY <= 0.002 mm, PERPENDICULAR TO [A] WITHIN 0.005 mm. SURFACE FINISH Ra <= 0.2 MICRONS TO MAINTAIN CHIP COPLANARITY ANGULAR DRIFT < 1.0 µrad UNDER 20G DYNAMIC IMPACT.
3.3 2X SYMMETRIC WEIGHT-REDUCTION POCKETS (32.0 x 8.0 mm x DEPTH 18.0 mm): 100% COAXIAL WITH MOUNTING BOLT PATTERN (X=10.0 & X=50.0 mm). MINIMUM WALL THICKNESS TO CENTRAL BORE >= 4.00 mm TO ELIMINATE STRUCTURAL CHATTER.
3.4 4X CORNER MOUNTING COUNTERBORES: CENTERED AT (10.0, 6.0) AND (50.0, 6.0) GRID, TRUE POSITION WITHIN Ø0.05 mm RELATIVE TO [A], [C], [D].

4. CLEANING & SURFACE FINISH:
4.1 DEBURR 100% UNDER 30X MAGNIFICATION. ALL SHARP EDGES BROKEN R0.20 mm OR C0.20 mm MAX.
4.2 SURFACE TREATMENT: FOR Ti-6Al-4V: HIGH-PRECISION ELECTROLESS NICKEL (ENP, MIL-C-26074, 5 µm); FOR AL 7075: HARD ANODIZE TYPE III (MIL-A-8625, 20-25 µm, PTFE SEALED).
4.3 CLEANROOM PACKAGING: ULTRASONICALLY CLEANED WITH 18 MOhm·cm DEIONIZED WATER, VACUUM SEALED FOR CLASS 1 / ISO 1 CLEANROOM INTEGRATION.





For reference

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.02 mm, Angular ±0.1° (Ultra-Precision 5-Axis Turn-Milling)		FINISH: For Ti-6Al-4V: Micro-Arc Oxidation (MAO / PEO, 3-5 µm) or Electroless Nickel Plating (MIL-C-26074, 5 µm) / For Al 7075: Hard Anodize Type III (MIL-A-8625, 20-25 µm, Ultra-Clean PTFE Sealed)		DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE		TITLE: FLIP-CHIP HIGH-SPEED BONDHEAD VOICE-COIL SUSPENSION MOUNT	
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: Option A: Vacuum-Melted Titanium Alloy Ti-6Al-4V ELI (Grade 23 / AMS 4930) / Option B: Forged Aerospace Aluminum 7075-T6 (AMS 4045)	DWG NO. W3-SEMI-FLIPCHIP-BONDHEAD-6.4	A3
				WEIGHT: ~333.4 g (Ti-6Al-4V ELI) / ~211.5 g (Al 7075-T6)	SCALE NTS	SHEET 1 OF 1